# PAIR-SV Train15 Interim Study
# Supplementary material

Author placeholder

## Scope and reproducibility boundary

This supplement documents the frozen Train15 / validation comparison. It contains 29,924 successful training rows from 15 people and 9,981 successful validation rows from five distinct people. The strict 50-person manifest contained 100,000 rows and a planned 40/5/5 person split. No internal-test person, input, prediction or metric was accessed. The controlled artifacts used seed 20260720 and the same row-aligned validation order for M0, M2 and M4.

## S1. Strict manifest and label rules

Positive labels required a sequence-resolved GRCh38 deletion of 50–10,000 bp, valid REF/ALT representation and coordinates, non-missing 0/1 or 1/1 genotype, GQ $\geq$ 20, and the selected PanGenie/HGSVC2 site-quality policy. The selected manifest retained GQ for QC only. Same-site negatives required a non-missing 0/0 genotype, GQ $\geq$ 20, callability, no overlapping conflicting carried deletion, and later adequate local read coverage. Positive sampling was deterministic and stratified by chromosome, deletion-length and allele-frequency bucket; matching used same candidate locus where possible, then continuous allele-frequency matching with length, chromosome and GQ as deterministic secondary criteria. Sample ID, population, split, GQ, allele frequency, negative type, source filename and read-name hash were excluded from model tensors.

Table 1: Table S1. Seventeen scalar read-support features.

| Feature | Definition |
|---|---|
| `left_soft_clip_count` | Reads with at least 10 soft-clipped bases at the left breakpoint. |
| `right_soft_clip_count` | Reads with at least 10 soft-clipped bases at the right breakpoint. |
| `split_read_count` | Selected reads carrying split or deletion-CIGAR evidence. |
| `supplementary_alignment_count` | Selected reads flagged supplementary. |
| `discordant_pair_count` | Selected reads with anomalous mate orientation or separation. |
| `reference_spanning_read_count` | Ordinary selected reads spanning the local reference interval. |
| `candidate_alt_support_count` | Selected reads prioritized as candidate-supporting. |
| `inside_depth` | Local depth over the deletion interior. |
| `left_flank_depth` | Depth in the left flanking window. |
| `right_flank_depth` | Depth in the right flanking window. |
| `inside_flank_depth_ratio` | Interior depth divided by mean flank depth with numerical guard. |
| `mean_mapq` | Mean mapping quality of selected reads. |
| `mean_base_quality` | Mean selected-read base quality. |
| `mean_abs_insert_size` | Mean absolute inferred insert size. |
| `insert_size_anomaly_fraction` | Fraction of selected reads with anomalous insert size. |
| `left_right_evidence_balance` | Normalized balance between left and right breakpoint evidence. |
| `log1p_deletion_length` | Natural log of one plus deletion length. |

## S2. Extraction container and QC

Validation extraction attempted 10,000 rows and completed 9,981 (0.19% failure). The explicit failure categories were 15 empty-read cases and four inadequate-flank-coverage cases; no failed row was silently zero-filled. Successful rows had finite scalars and non-empty image/read tensors. The validator checked all five validation people, GQ minimum 20, balanced planned classes, 3,412 same-candidate positive/negative pairs with identical coordinates and

Table 2: Table S2. Training-ready shard schema.

| Item | Shape / type | Purpose |
|---|---|---|
| Pileup images | $(2, 6, 64, 128)$ uint8 | Left/right breakpoint images |
| Read tokens | $(2, 48, 128)$ uint8 | Sample-specific selected sequences |
| Read mask | $(2, 48)$ uint8 | Padding-aware pooling |
| Scalars | (17) float32 | Read-support summary vector |
| Candidate REF/ALT | $(2, 1024)$ uint8, deduplicated | Candidate-level GRCh38 context |
| Row metadata | IDs, label, coordinates, success/failure | QC and alignment only |
| Read metadata | CIGAR, strand, MAPQ, BQ, flags, reason | QC only; read-name hash excluded |

sample-specific evidence differences, plus the model-tensor whitelist. The HDF5 schema was `pairsv-shard-v1`; the validation HDF5 SHA-256 was `af409617a3959ed8eaadb66039081d9ef80cfda84d089920fe306e5df774362a`.

# S3. Controlled training protocol

Table 3: Table S3. Fixed Train15 training configuration.

| Item | Fixed value |
|---|---|
| Seed | 20260720 |
| Train / validation people | 15 / 5 |
| Successful rows | 29,924 / 9,981 |
| Epochs | 5 maximum; selected checkpoint by validation F1 |
| Optimizer | AdamW |
| Learning rate / weight decay | $2 \times 10^{-4}$ / $10^{-4}$ |
| Batch size | 16 |
| Loss / precision | Cross-entropy / mixed precision |
| Normalization | Fit only on training people |
| Decision threshold | Maximize validation F1 per model |
| M0 / M2 / M4 parameters | 287,618 / 323,110 / 622,374 |

# S4. Validation subgroups and per-person results

Table 4: Table S4. Validation subgroups from frozen row-aligned predictions.

| Model | Subgroup | $n$ | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| M0 | 50–200 bp | 6,781 | .8211 | .7740 | .8623 |
| M2 | 50–200 bp | 6,781 | .8193 | .7733 | .8632 |
| M4 | 50–200 bp | 6,781 | .8344 | .7832 | .8716 |
| M0 | 200–1,000 bp | 2,367 | .8963 | .8430 | .9185 |
| M2 | 200–1,000 bp | 2,367 | .8931 | .8419 | .9184 |
| M4 | 200–1,000 bp | 2,367 | .8882 | .8477 | .9226 |
| M0 | 1,000–10,000 bp | 833 | .8404 | .8718 | .9228 |
| M2 | 1,000–10,000 bp | 833 | .8607 | .8646 | .9250 |
| M4 | 1,000–10,000 bp | 833 | .8584 | .8692 | .9331 |
| M0 | GT 0/1 | 3,818 | .8143 | .8976 | – |
| M2 | GT 0/1 | 3,818 | .8169 | .8992 | – |
| M4 | GT 0/1 | 3,818 | .8214 | .9019 | – |
| M0 | GT 1/1 | 996 | .9378 | .9679 | – |
| M2 | GT 1/1 | 996 | .9247 | .9609 | – |
| M4 | GT 1/1 | 996 | .9518 | .9753 | – |

Table 5: Table S5. Per-person validation F1. Counts are successful rows.

| Person | $n$ | M0 F1 | M2 F1 | M4 F1 |
|---|---|---|---|---|
| HG00106 | 1,999 | .8080 | .8025 | .8057 |
| HG00422 | 1,992 | .7786 | .7806 | .7843 |
| HG00732 | 1,995 | .8085 | .8083 | .8235 |
| HG01894 | 1,998 | .7913 | .7905 | .7992 |
| HG02597 | 1,997 | .8084 | .8072 | .8174 |

Same-site negatives are a negative-only stratum, so positive-class recall/F1 are not defined there. Their classification error is represented in the main-paper confusion matrices; for M4, TN/FP is 3,708/1,288 at its validation-selected threshold.

## S5. Branch interventions and uncertainty

Table 6: Table S6. M4 branch interventions.

| Intervention | F1 | ΔF1 | ΔROC-AUC | ΔPR-AUC | Changed predictions |
|---|---|---|---|---|---|
| Candidate zero | .7998 | -.0063 | -.0020 | -.0015 | 4.86% |
| Read zero | .7980 | -.0082 | -.0082 | -.0065 | 5.49% |
| Scalar zero | .7905 | -.0156 | -.0158 | -.0161 | 8.15% |
| Candidate shuffle | .8003 | -.0058 | -.0064 | -.0056 | 5.55% |
| Read shuffle | .8120 | +.0058 | +.0039 | +.0035 | 7.70% |
| Scalar shuffle | .8415 | +.0354 | +.0333 | +.0288 | 11.50% |

Zero ablation is a functional dependence check but out of distribution. Shuffle intervention breaks row correspondence and was unstable in this finite validation realization; it is therefore reported but not interpreted as causal feature importance.

The paired bootstrap resampled complete aligned sample–candidate rows 1,000 times. It estimated M4–M0 F1 +0.00697 (95% CI +0.00164 to +0.01238), M4–M2 +0.00815 (+0.00292 to +0.01381), and M2–M0 −0.00096 (−0.00499 to +0.00337). Rows can reuse candidate loci across people; therefore this is not a candidate-cluster bootstrap and the interval must not be read as fully independent-locus uncertainty.

## S6. Checksums and remaining limitations

The controlled metrics JSON SHA-256 is `8af000e4074f6e15e1cb0fc30c714d207d93d8dc68173e02251025f17a669b93`; aligned predictions CSV SHA-256 is `4617dfb2d6922a6a7dd56469785ead01667416240cfa81c14217bb7ff9844f52`; and ablations CSV SHA-256 is `bc4370aaa49bd05ac328eaf8a6f208dcccad9d7bf4a8a49e5d3ef6298be12b07`. The study remains limited by one seed, 15 training people, validation-selected thresholds, silver labels, selected-candidate classification rather than discovery, residual locus dependence and row-level uncertainty. Final claims require full 40-person training, three seeds, candidate-cluster uncertainty and a single untouched internal-test evaluation.